THE THICK SUBCATEGORY GENERATED BY ONE OBJECT

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 11204-11227, following the statement at 11191-11202.

Editorial note. No omissions: both closure and minimality are argued with the four operations listed.

Allowed standard prerequisites (editorial).

- Explicit description of the generated subcategory. lemma-generated-by-E-explicit (same chapter, cited at ~11205 and ~11212). Gives the union formula and the extension index bound <E>_a + <E>_b subset <E>_{a+b}.

Source reference. Original lemma 0FX9: sources/stacks-derived/pinned-original-source.tex, source lines 11166–11166. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{D}$ be a triangulated category. Let E be an object of $\mathcal{D}$. The subcategory*

$$\langle E \rangle = \bigcup\nolimits_n \langle E \rangle_n = \bigcup\nolimits_{n,m \geq 1} smd(add(E[-m, m])^{\star n})$$

is a strictly full, saturated, triangulated subcategory of $\mathcal{D}$ and it is the smallest such subcategory of $\mathcal{D}$ containing the object E.

Proof. The equality on the right follows from Lemma 0FX9. It is clear that $\langle E \rangle = \bigcup \langle E \rangle_n$ contains E, is preserved under shifts, direct sums, direct summands. If $A \in \langle E \rangle_a$ and $B \in \langle E \rangle_b$ and if $A \to X \to B \to A[1]$ is a distinguished triangle, then $X \in \langle E \rangle_{a+b}$ by Lemma 0FX9. Hence $\bigcup \langle E \rangle_n$ is also preserved under extensions and it follows that it is a triangulated subcategory.

Finally, let $\mathcal{D}' \subset \mathcal{D}$ be a strictly full, saturated, triangulated subcategory of $\mathcal{D}$ containing E. Then $\mathcal{D}'[-\infty, \infty] \subset \mathcal{D}'$, $add(\mathcal{D}) \subset \mathcal{D}'$, $smd(\mathcal{D}') \subset \mathcal{D}'$, and $\mathcal{D}' \star \mathcal{D}' \subset \mathcal{D}'$. In other words, all the operations we used to construct $\langle E \rangle$ out of E preserve $\mathcal{D}'$. Hence $\langle E \rangle \subset \mathcal{D}'$ and this finishes the proof. □

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